Shreyas Suman

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Education

Santa Clara University, School of Engineering — Santa Clara, CA
Bachelor of Science in Computer Science & Engineering — *Expected June 2030*

Relevant Coursework: Advanced Programming (C), Data Structures, Discrete Mathematics, Calculus III

Experience

Software Engineer Intern — May 2025 – Sept. 2025
Stadea Tools

- Automated weekly sales forecasting for a retail store's purchasing decisions, processing 1,000+ transactions per week, by building the Java backend and an Apache Commons IO ETL pipeline
- Shipped 5+ production deployments by collaborating on feature design and code reviews, adding error handling and retry logic so weekly data loads ran reliably
- Secured the store's adoption of the tool for weekly purchasing by presenting forecasting results to stakeholders

Leadership & Activities

Head of Computer Science Department — Dec. 2024 – Jan. 2026
Stellar Learning (Student-Run Nonprofit, Free AP & SAT Prep)

- Reached 25,000+ enrolled students by scaling the CS department from 4 to 9 free courses, more than doubling the catalog
- Grew a 10+ person volunteer instructor team over a 14-month term by owning recruiting, onboarding, and training
- Enabled new instructors to teach from day one across all 9 courses by designing a standardized curriculum framework
- Delivered 200+ live online sessions to students across multiple time zones by directing scheduling and operations

Projects

TalosAI | *TypeScript, Python, Supabase, React* — **3rd Place**, CrowdStrike x AIC Hackathon – Oct. 2026

- Built and deployed the TypeScript backend on Supabase Edge Functions, serving 3 AI triage tools that turn security alerts, emails, and malware types into plain-English verdicts
- Caught 98% of malicious emails (147/150) at a 4.7% false-alarm rate on 300 held-out emails by building a pipeline that scores threat signals in code before the AI, validated over 6 rounds on 1,500 emails
- Led the malware module, building a Python data pipeline that turns CIC-MalMem-2022 forensics data into profiles for 3 malware families

Spectra | *C++, JUCE*

- Released an open-source spectral freeze plugin (VST3/AU) that turns sound into a pad using a C++ FFT engine
- Ran every effect at once on 1.8% of one CPU core by keeping the audio thread free of locks and memory allocation
- Held pitch error under 1 Hz across every release by writing an automated test suite that checks the audio offline

Sleeper | *Python, FastAPI, React, SQLite*

- Shipped a live album recommender that turns 5 favorite albums into 1 pick and 4 runners-up by building a FastAPI backend on Render and a React frontend on Vercel
- Raised album hit rate from 35% to 45% across 60 leave-one-out tests by adding per-artist popularity scoring
- Stayed under the request limits of 2 external APIs (Last.fm, MusicBrainz) by building a SQLite cache and rate limiter

Technical Skills

Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks & Libraries: React, FastAPI, JUCE, pytest, Apache Commons IO
Tools & Concepts: Git, CMake, GitHub Actions, SQLite, REST APIs, ETL, Real-Time Audio/DSP